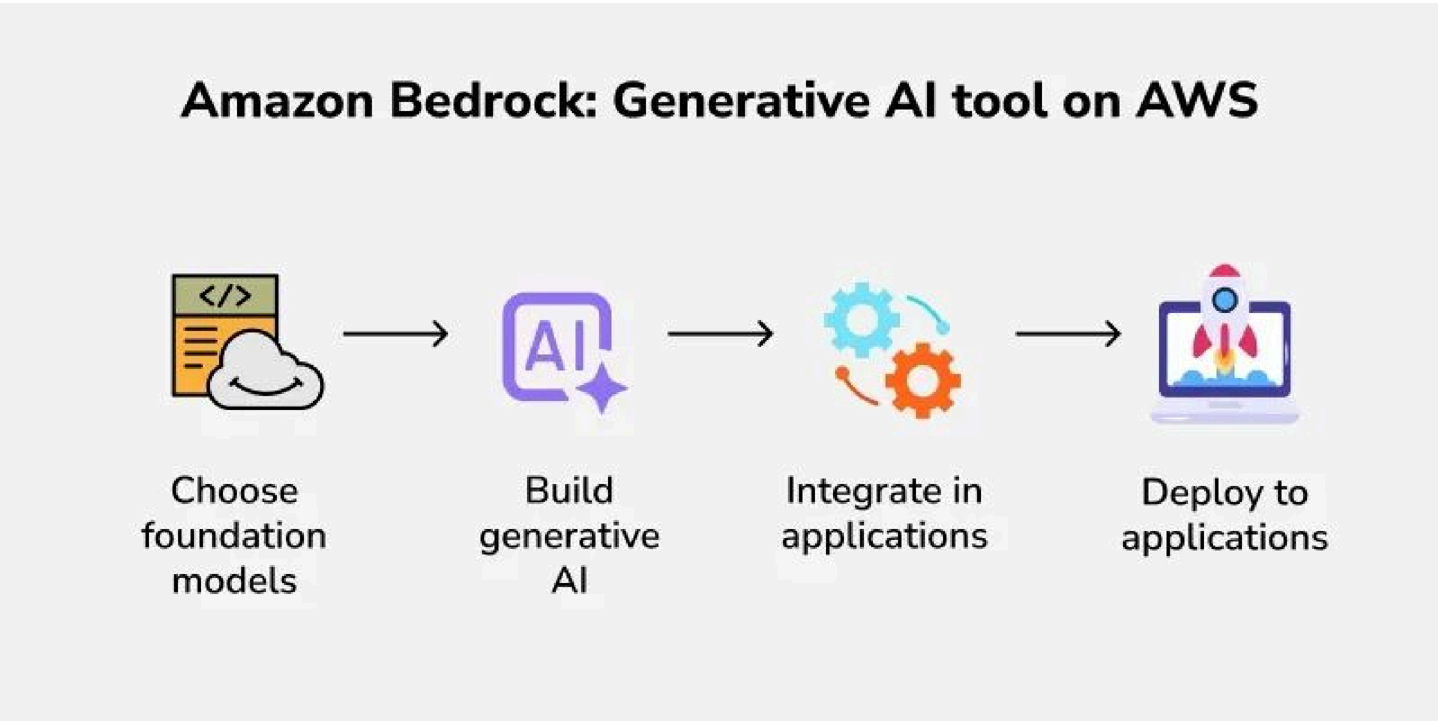
# *AWS AI Implementation Guide: Amazon Bedrock vs. Amazon SageMaker Analysis*

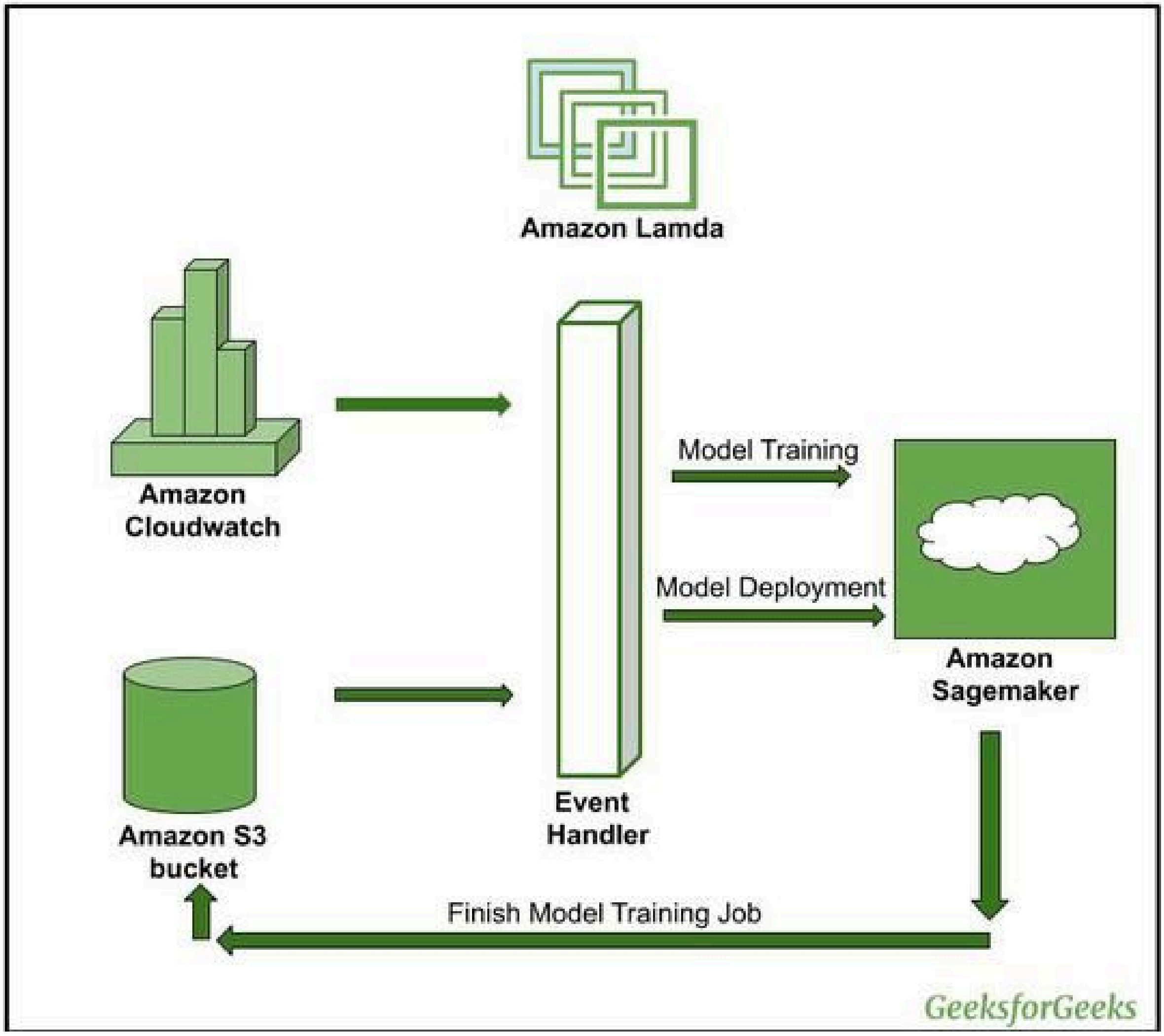
## *Amazon Bedrock*

is a fully managed, serverless service designed for **quickly integrating generative AI** into applications using pre-trained foundation models (e.g., Amazon Titan, Anthropic Claude, AI21, Stability AI) via a simple API. It's ideal for developers and non-ML experts who want fast prototyping, minimal setup, and cost efficiency for sporadic or variable inference workloads.




## *Amazon SageMaker*

is a comprehensive, end-to-end machine learning platform for building, training, and deploying **custom ML and deep learning models at scale**. It supports full lifecycle workflows including data preprocessing, model training, hyperparameter tuning, deployment, and monitoring. SageMaker offers granular control over infrastructure, algorithms, and model customization—making it best suited for data scientists, ML engineers, and teams with advanced ML expertise.




## The Critical Differences

| Feature | Amazon Bedrock | Amazon SageMaker |
|---|---|---|
| **Primary User** | App Developers / Product Teams | Data Scientists / ML Engineers |
| **Effort / Setup** | Near Zero (API-driven) | Significant (Requires Infrastructure Setup) |
| **Pricing Model** | **Pay-per-token:** High variable cost, zero fixed cost. | **Pay-per-instance:** Fixed hourly cost for running servers. |
| **Customization** | Fine-tuning (Limited to specific models) | Full training, Hyperparameter tuning, RLHF. |
| **Privacy/VPC** | Data is encrypted; stays within AWS. | Can be deployed 100% within a private VPC. |
| **Latency** | Variable (Shared infrastructure) | Guaranteed/Low (Dedicated hardware) |

## When to Choose Which?

### <u>Choose Amazon Bedrock if:</u>

- You need to launch a **Minimum Viable Product (MVP)** in days.
- Your workload is **sporadic** (e.g., a chatbot used occasionally).
- Your team has **limited ML expertise**.

### <u>Choose Amazon SageMaker if:</u>

- You have **high-volume traffic** (e.g., 200M+ tokens/day). At this scale, renting a dedicated server in SageMaker is often cheaper than paying per token in Bedrock.
- You need **sub-50ms latency** for real-time applications.
- You are doing **Non-Generative AI** (Predicting house prices, churn analysis, or fraud detection).